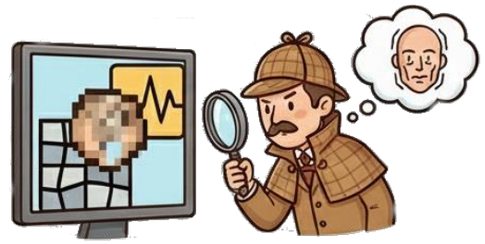
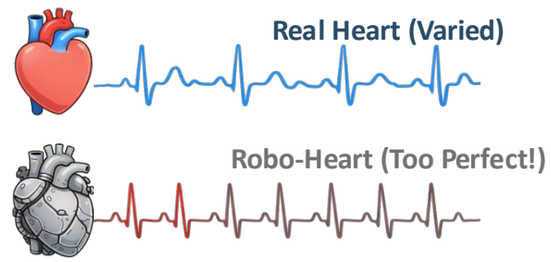
Visual View

Layer 1 Intrinsic Cues Analysis (Section 4.1)

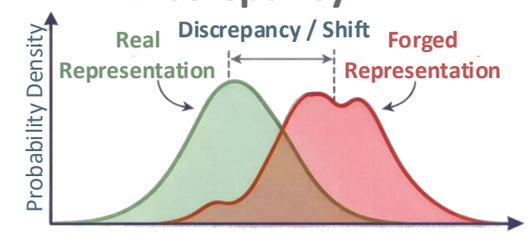
1. Pixel & Geometric Artifacts



2. Physiological Features

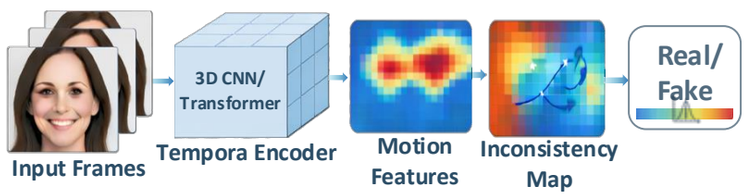


3. Representation Distribution Discrepancy

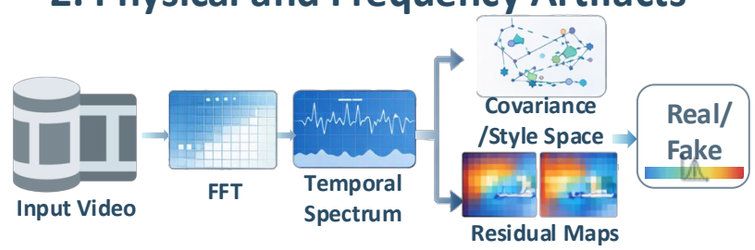


Layer 2 Spatiotemporal Consistency (Section 4.2)

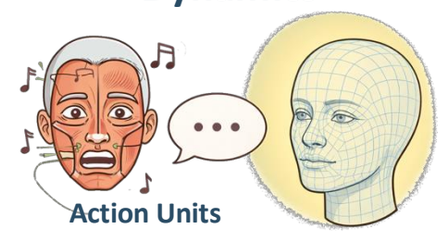
1. Temporal and Motion Inconsistencies



2. Physical and Frequency Artifacts



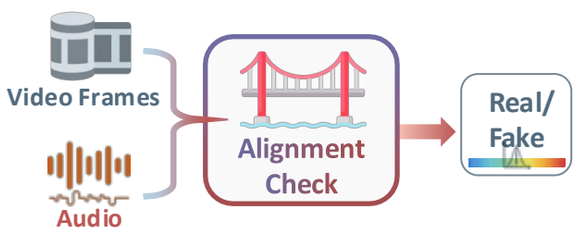
3. Human Behavioral and Interaction Dynamics



Language View

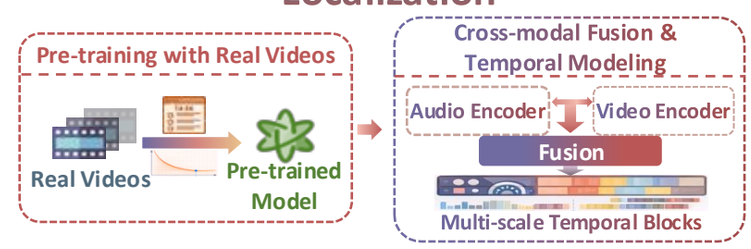
Layer 3 Cross-Modal Consistency (Section 4.3)

1. Audio-Visual Consistency Detection



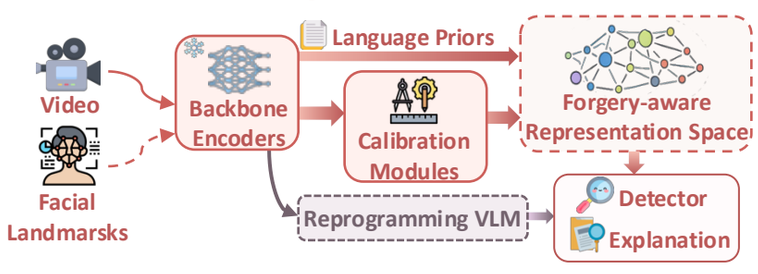
2. Text-Visual Semantics Consistency Reasoning

3. Robust Learning and Temporal Localization

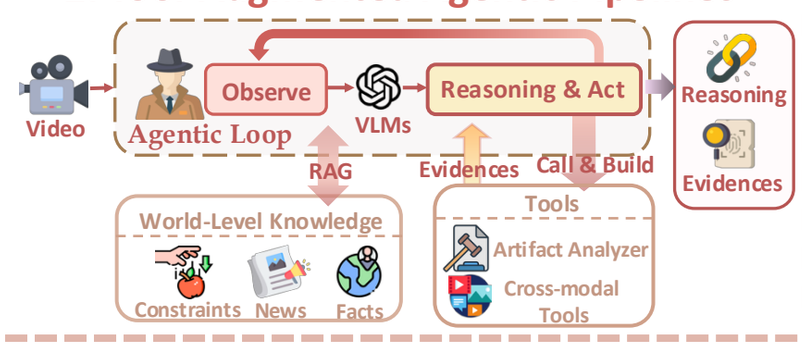
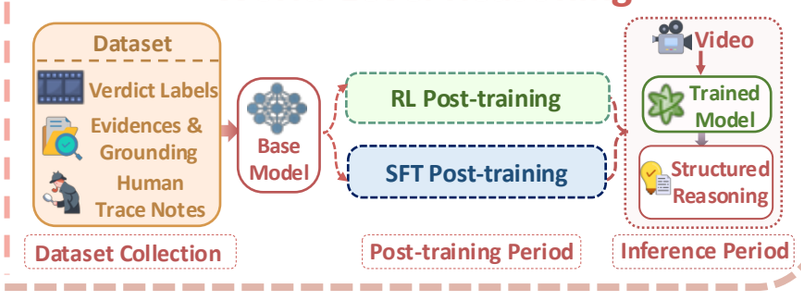


Layer 4 Language-Guided World-Level Reasoning (Section 4.4)

1. Language-Calibrated Forgery Representation



2. Tool-Augmented Agentic Pipelines



3. Post-Training for Explainable World-Level Reasoning